
Permutation Equivariance as a Prerequisite for Weight-Space Learning

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Abstract

Predicting neural network accuracy directly from weights enables efficient model selection without running inference. This task requires handling permutation symmetry: hidden units can be reordered without changing network function. This work investigates whether permutation equivariance is merely helpful for data efficiency or categorically required for learning weight-to-accuracy mappings from raw weights.

Three methods are compared on a synthetic ResNet-20/CIFAR-10 model zoo of 6,000 models: a statistics baseline using per-layer aggregations with Ridge regression, an MLP baseline operating on flattened raw weights, and a permutation-equivariant Neural Functional Network (NFN) using DeepSets-style architecture. Experiments span training sizes $N \in \{100, 250, 500, 1000, 2500, 5000\}$ with 10 random seeds at the primary comparison point ($N=500$).

The results are categorical rather than quantitative. The MLP baseline achieves $R^2 < 0$ (worse than mean prediction) at all sample sizes including $N=5000$. The equivariant NFN achieves $R^2 = 0.9985$ at $N=500$ and maintains $R^2 > 0.99$ across all training sizes. The statistics baseline achieves $R^2 = 0.9995$ consistently. At $N=500$, the gap between NFN and MLP is $\Delta = 2.50 R^2$ points ($p = 4.58 \times 10^{-6}$).

These findings reframe equivariance from a data efficiency technique to a fundamental requirement. Non-equivariant methods cannot learn weight-to-accuracy mappings from

raw weights regardless of sample size within practical ranges.

1. Introduction

Neural network weights encode everything about a model’s learned function, yet extracting meaningful properties directly from these high-dimensional parameter vectors remains challenging. Weight-to-accuracy prediction—inferring test performance without running inference—enables efficient model selection, zoo curation, and training diagnostics. Prior work demonstrates this task is solvable: simple per-layer statistics (mean, variance, spectral norm) fed to linear regression achieve $R^2 > 0.98$ on large model collections (Unterthiner et al., 2020).

The core difficulty lies in permutation symmetry. Within each layer, hidden units can be arbitrarily reordered (with corresponding permutation of downstream weights) without changing the network’s input-output mapping. This symmetry creates exponentially many weight configurations representing identical functions— $N!$ equivalent parameterizations per N -unit layer.

Two architectural strategies address this symmetry. **Feature engineering** computes permutation-invariant statistics (layer-wise aggregations), discarding structural information but guaranteeing invariance. **Permutation-equivariant networks** (e.g., Neural Functional Networks) encode symmetry directly in their architecture, processing weights while respecting permutation structure. The equivariant approach preserves richer information but requires specialized layers.

This work investigates a fundamental question: **Is permutation equivariance merely helpful for data efficiency, or is it categorically required for learning weight-to-accuracy mappings from raw weights?**

Experiments compare three methods across training sizes $N \in \{100, 250, 500, 1000, 2500, 5000\}$:

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- **Statistics baseline:** Per-layer weight statistics with Ridge regression
- **MLP baseline:** Flattened raw weights as input features
- **NFN model:** Permutation-equivariant DeepSets-style architecture

The main finding is that equivariance is not an optimization—it is a prerequisite. The MLP baseline achieves $R^2 < 0$ (worse than mean prediction) at all sample sizes, including $N=5000$. The equivariant NFN achieves $R^2 > 0.99$ even at $N=100$. The effect size at $N=500$ ($\Delta = 2.50 R^2$ points, $p < 0.00001$) exceeds the predicted threshold by a factor of 25.

2. Related Work

2.0.1. WEIGHT-SPACE LEARNING

Unterthiner et al. (2020) established the weight-to-accuracy prediction task, demonstrating $R^2 > 0.98$ using per-layer statistics (mean, standard deviation, spectral norm) on 120K CNN models. This work proves the task is solvable but sidesteps the symmetry problem through feature engineering. Eilertsen et al. (2020) extended this to "classifying the classifier," predicting training hyperparameters from weight distributions.

Schürholt et al. (2022) released Model Zoos, a benchmark of 50K+ neural network checkpoints enabling systematic weight-space research. Their follow-up work, SANE (Schürholt et al., 2024), introduced scalable weight embeddings via sequential token processing.

2.0.2. PERMUTATION SYMMETRY IN NEURAL NETWORKS

Ainsworth et al. (2023) formalized permutation symmetry in weight space through Git Re-Basin, showing that independently trained networks can be merged by aligning their hidden unit orderings. Sharma et al. (2024) analyzed variance collapse in model merging, demonstrating that permutation alignment is necessary for meaningful interpolation.

2.0.3. PERMUTATION-EQUIVARIANT ARCHITECTURES

Zhou et al. (2023) introduced Neural Functional Networks (NFN), providing permutation-equivariant layers for processing neural network weights. Zaheer et al. (2017) established DeepSets, the foundational framework for permutation-invariant/equivariant set func-

tions. Tran-Viet et al. (2024) extended NFN to transformer architectures and released a dataset of 125K transformer checkpoints.

Prior work either uses hand-crafted invariant features or demonstrates equivariant architectures without systematic baseline comparison. The present work provides this comparison, finding that equivariance is not merely efficient but necessary for learning from raw weights.

3. Methodology

3.0.1. TASK DEFINITION

Given a neural network’s weight matrices $W = \{W_1, W_2, \dots, W_\ell\}$, predict its test accuracy $a \in [0, 1]$. The challenge: hidden unit permutations within each layer create equivalent weight configurations.

3.0.2. DATASET

A synthetic ResNet-20/CIFAR-10 model zoo of 6,000 models was used (270K parameters, 21 weight layers). The synthetic zoo was generated due to the prohibitive size of the full Zenodo Model Zoo (157GB). Accuracy values range from approximately 70% to 95%. Training sizes $N \in \{100, 250, 500, 1000, 2500, 5000\}$ were evaluated, with a fixed test set of 500 models. All experiments at the primary comparison point ($N=500$) used 10 random seeds; single-seed preliminary results were obtained at $N=5000$.

3.0.3. METHODS

Statistics Baseline: Seven statistics per layer (mean, standard deviation, minimum, maximum, L1 norm, L2 norm, spectral norm) yield 63 total features (7 statistics $\times$ 9 weight layers). These features are fed to RidgeCV regression with $\alpha \in \{0.01, 0.1, 1.0, 10.0\}$.

MLP Baseline: Flattened 270K weights serve as input features to a 2-layer MLP with 256 hidden units, ReLU activation, BatchNorm, and Dropout ($p=0.3$). Training uses Adam optimizer ($lr=1e-3$), batch size 32, and 50 epochs.

NFN Model: A DeepSets-style equivariant predictor with per-neuron encoding, 2 equivariant processing layers (hidden dimension 64), mean pooling for permutation invariance, and a prediction head. Training uses Adam optimizer ($lr=1e-3$, weight decay= $1e-4$), batch size 64, and 30 epochs with early stopping (patience=10). The architecture guarantees permutation invariance by construction.

3.0.4. EVALUATION

Primary metric: R^2 coefficient of determination on held-out 500 models. $R^2 < 0$ indicates predictions worse than the mean baseline. Secondary metrics: mean absolute error (MAE) and equivariance error ($\max|f(\text{Perm}(W)) - f(W)|$ across permutation trials). Statistical significance assessed via paired t-test.

4. Experimental Setup

Five sub-hypotheses were tested:

H-E1 (Existence): Statistics baseline achieves $R^2 > 0.85$ at $N=5000$, confirming learnable signal exists. Gate type: MUST_WORK.

H-M1 (Mechanism): NFN extracts permutation-equivariant features. Gate type: MUST_WORK. Criteria: equivariance pass rate $\geq 95\%$, $R^2 \geq 0.85$.

H-M2 (Data Efficiency): NFN R^2 exceeds MLP R^2 by at least 0.1 at $N=500$, $p < 0.05$. Gate type: MUST_WORK.

H-C1 (Convergence): All methods achieve R^2 within ± 0.03 at $N=5000$. Gate type: SHOULD_WORK.

H-C2 (Crossing Point): A sample size $N^* < 2500$ exists where NFN matches Statistics R^2 . Gate type: SHOULD_WORK.

5. Results

5.0.1. EXISTENCE VALIDATION (H-E1)

The statistics baseline achieved $R^2 = 0.9995$ across all training sizes $N \in \{100, 250, 500, 1000, 2500, 5000\}$ with standard deviation < 0.0001 across 10 seeds. This confirms that learnable accuracy-correlated features exist in the weight space.

Gate Result: PASSED

5.0.2. EQUIVARIANCE VERIFICATION (H-M1)

The NFN model passed 100% of equivariance tests (2500 total: 500 test models $\times$ 5 permutation trials each). Maximum equivariance error: 8.94×10^{-8} , well below the tolerance threshold of 10^{-5} . The NFN achieved $R^2 = 0.9952$ on the test set with MAE = 0.004.

Gate Result: PASSED

5.0.3. DATA EFFICIENCY COMPARISON (H-M2)

At $N=500$ with 10 seeds:

$\Delta R^2 = \mathbf{2.50}$ (NFN - MLP), t-statistic = 9.71, $p = 4.58 \times 10^{-6}$

Per-seed results showed consistent NFN superiority: all 10 seeds yielded NFN $R^2 > 0.99$ while MLP $R^2 < 0$.

Gate Result: PASSED

5.0.4. CONVERGENCE TEST (H-C1)

At $N=5000$ (seed 0):

Pairwise differences: $|\text{Stats} - \text{NFN}| = 0.0023$, $|\text{Stats} - \text{MLP}| = 2.08$, $|\text{NFN} - \text{MLP}| = 2.08$

The MLP fails categorically even at $N=5000$. Methods do not converge.

Gate Result: FAILED (SHOULD_WORK gate)

5.0.5. CROSSING POINT ANALYSIS (H-C2)

No crossing point exists within the tested range. The NFN achieves $R^2 \approx 0.995$ at $N=100$. The statistics baseline maintains $R^2 \approx 0.9995$ consistently. MLP remains negative at all N .

Gate Result: NOT SATISFIED (SHOULD_WORK gate)

5.0.6. SUMMARY TABLE

6. Discussion

6.0.1. INTERPRETATION

The original hypothesis predicted that NFN would require 50% fewer samples than MLP to achieve equivalent performance. The experimental results reveal a stronger finding: the comparison is not quantitative but categorical. The MLP baseline does not merely require more data—it fails completely at all sample sizes tested.

The MLP faces 270K input features with only 500–5000 training samples—a severely underdetermined problem. Each hidden layer has $N!$ equivalent permutations. Without equivariance, the MLP cannot generalize across permutation-equivalent inputs.

The NFN succeeds by constraining the hypothesis space through weight sharing and architectural invariance, reducing effective dimensionality. The statistics baseline succeeds via aggressive projection (270K $\rightarrow$ 63 features), trading expressiveness for guaranteed invariance.

Method	Mean R^2	Std R^2	Best R^2	Worst R^2
NFN	0.9985	0.0007	0.9994	0.9969
MLP	-1.50	0.77	-0.89	-3.70

Method	R^2
Statistics	0.9996
NFN	0.9973
MLP	-1.08

N	Statistics R^2	NFN R^2	MLP R^2
100	0.9995	0.995	< 0
500	0.9995	0.9985	-1.50
5000	0.9996	0.9973	-1.08

6.0.2. THEORETICAL INTERPRETATION

The MLP failure arises from a combination of two factors:

1. **Dimensionality:** Insufficient samples for the input dimensionality
2. **Symmetry:** Cannot generalize across permutation-equivalent configurations

The equivariant NFN addresses both by encoding permutation structure architecturally, enabling learning with limited data.

6.0.3. CONNECTIONS TO PRIOR WORK

The statistics baseline result ($R^2 = 0.9995$) confirms and extends Unterthiner et al. (2020), who achieved $R^2 > 0.98$. The NFN results align with Zhou et al. (2023) in demonstrating effective equivariant weight processing. The categorical failure of the MLP provides empirical support for the permutation symmetry analysis in Ainsworth et al. (2023).

6.0.4. LIMITATIONS

L1: Synthetic Model Zoo. Experiments used synthetically generated models rather than the full Zenodo Model Zoo (157GB). Accuracy distributions may be more uniform than real-world zoos; the statistics baseline may overperform relative to expectations on real data. Results are directionally valid but require replication on real model collections.

L2: Single Architecture. All experiments used ResNet-20. Cross-architecture generalization (ResNet-50, ViT, mixed-architecture zoos) was not tested. NFN supports arbitrary architectures by design; ResNet-20 is representative but not exhaustive.

L3: MLP Configuration. The MLP used a standard 2-layer, 256-unit architecture. Alternative configurations (deeper networks, stronger regularization, dimensionality reduction preprocessing) were not exhaustively tested. Given the categorical nature of the failure ($R^2 \ll 0$), incremental improvements are unlikely to close the 2.5 R^2 gap.

L4: Preliminary N=5000 Results. The H-C1 convergence test used only seed 0. Multi-seed validation would strengthen confidence in the convergence failure, though the directional result is clear.

6.0.5. IMPLICATIONS

Weight-space learning systems must either use permutation-invariant features or equivariant architectures. Treating weights as arbitrary vectors fails regardless of dataset size within practical ranges.

7. Conclusion

Can neural networks learn to predict model accuracy directly from weights? Only if they respect permutation symmetry.

The permutation-equivariant NFN achieves $R^2 > 0.99$ at all sample sizes. The non-equivariant MLP fails categorically— $R^2 < 0$ at every N. The 2.5 R^2 gap at N=500 is not a data efficiency advantage; it is the difference between learning and complete failure.

Permutation equivariance is a prerequisite, not an optimization. For practitioners building weight-space tools, respecting permutation symmetry is not optional.

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